

HALLANDSASEN, Sweden

WMO: 026020

Lat: 56.4040N

Lon: 12.9436E

Elev: 159

StdP: 99.43

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.1	-14.8	1.1	-5.7	-13.2	1.2	-0.4	6.5	1.3	5.9	1.3	0.8	230	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	25.9	18.3	24.1	17.4	22.4	16.4	19.6	23.8	18.6	22.4	17.7	21.1	1.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.3	21.3	17.2	12.5	20.3	16.2	11.7	19.6	56.5	23.8	53.4	22.4	50.4	21.2	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.5	4.0	-15.7	28.5	2.4	1.9	-17.5	29.9	-18.9	31.0	-20.3	32.0	-22.1	33.4		
			-16.0	21.2	2.3	1.2	-17.6	22.1	-18.9	22.8	-20.2	23.5	-21.8	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-1.1	-0.5	1.9	6.6	10.7	13.9	16.6	16.2	12.3	8.1	4.3	0.7
	DBStd	7.13	4.44	3.86	3.87	3.54	3.47	2.77	2.59	2.55	2.78	3.53	3.69	4.62
	HDD10.0	1582	345	294	252	112	31	2	0	0	9	78	171	287
	HDD18.3	3973	603	528	510	351	237	137	68	77	180	318	420	546
	CDD10.0	678	0	0	1	11	53	118	204	192	80	18	1	0
	CDD18.3	29	0	0	0	0	1	3	14	11	1	0	0	0
	CDH23.3	226	0	0	0	0	14	41	120	51	1	0	0	0
	CDH26.7	22	0	0	0	0	1	3	16	2	0	0	0	0
Wind	WSAvg	1.6	2.0	1.8	1.7	1.6	1.6	1.3	1.2	1.1	1.3	1.7	2.0	1.8
Precipitation	PrecAvg	818	64	51	45	40	56	82	89	99	77	78	67	69
	PrecMax	1103	156	128	101	85	122	162	244	195	137	152	124	136
	PrecMin	570	3	6	16	5	20	0	3	24	17	31	18	14
	PrecStd	129	37	25	22	23	27	44	58	44	34	33	28	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.0	14.8	21.0	25.1	26.7	28.6	26.6	22.4	16.7	12.2	9.7
		MCWB	7.2	6.9	9.2	13.5	17.4	18.4	20.1	17.6	15.0	12.6	11.6	9.2
	2%	DB	6.6	6.5	10.8	17.8	22.1	24.4	26.4	24.7	20.0	14.7	11.1	8.5
		MCWB	5.8	5.4	7.3	11.1	14.8	16.9	18.9	17.9	15.1	12.0	10.4	7.9
	5%	DB	5.6	5.5	8.1	15.5	19.7	22.4	24.6	22.8	18.3	13.7	10.0	7.3
		MCWB	4.9	4.7	5.7	9.6	13.5	15.8	18.2	17.2	13.7	11.6	9.2	6.6
	10%	DB	4.6	4.5	6.7	13.1	17.1	20.0	22.7	20.9	16.8	12.8	9.1	6.4
		MCWB	3.9	3.7	5.2	8.2	11.9	14.3	17.1	16.3	13.0	11.0	8.3	5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	7.0	9.3	14.0	18.4	19.6	21.5	20.9	17.3	14.2	12.0	9.2
		MCDB	7.8	7.8	13.6	19.9	23.9	25.3	26.1	24.2	20.2	15.1	12.1	9.5
	2%	WB	6.0	5.7	7.4	11.6	16.2	17.7	20.0	19.1	15.9	12.9	10.5	8.0
		MCDB	6.5	6.2	9.9	16.8	20.2	22.7	24.7	22.2	19.0	14.1	10.9	8.4
	5%	WB	5.0	4.7	6.3	10.2	14.4	16.6	19.0	18.3	14.9	12.1	9.4	6.7
		MCDB	5.5	5.3	7.7	14.3	18.3	20.8	23.2	21.3	17.2	13.4	9.9	7.2
	10%	WB	4.0	3.7	5.2	8.8	12.8	15.3	18.0	17.4	14.0	11.3	8.4	5.8
		MCDB	4.5	4.4	6.5	12.2	16.2	19.0	21.5	20.2	15.9	12.5	9.0	6.3
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.4	6.7	9.6	9.5	9.4	9.1	8.2	7.4	6.0	4.1	3.9
		MCDBR	3.7	3.9	9.1	13.1	13.4	13.8	13.3	10.9	9.1	6.1	4.1	3.4
		MCWBR	3.8	3.4	6.1	7.7	7.9	7.6	7.1	5.8	5.4	4.3	3.7	3.6
	5% WB	MCDBR	3.6	3.7	7.8	12.0	11.7	11.9	11.2	9.4	8.5	5.6	3.9	3.6
		MCWBR	3.7	3.3	5.6	7.5	7.6	7.4	6.7	5.6	5.5	4.4	3.7	3.7
Clear-Sky Solar Irradiance	taub	0.300	0.325	0.345	0.379	0.377	0.369	0.391	0.391	0.360	0.346	0.325	0.292	
	taud	2.276	2.313	2.342	2.310	2.376	2.421	2.374	2.379	2.450	2.413	2.331	2.251	
	Ebn at Noon	592	731	814	837	863	874	843	816	793	701	557	498	
	Edh at Noon	52	75	93	111	110	107	110	103	84	68	50	42	
All-Sky Solar Radiation	RadAvg	0.41	1.00	2.36	3.97	5.18	5.44	5.05	4.06	2.86	1.37	0.52	0.28	
	RadStd	0.08	0.16	0.24	0.53	0.50	0.47	0.61	0.35	0.33	0.21	0.09	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-162	N/A	N/A

Nomenclature: See separate page